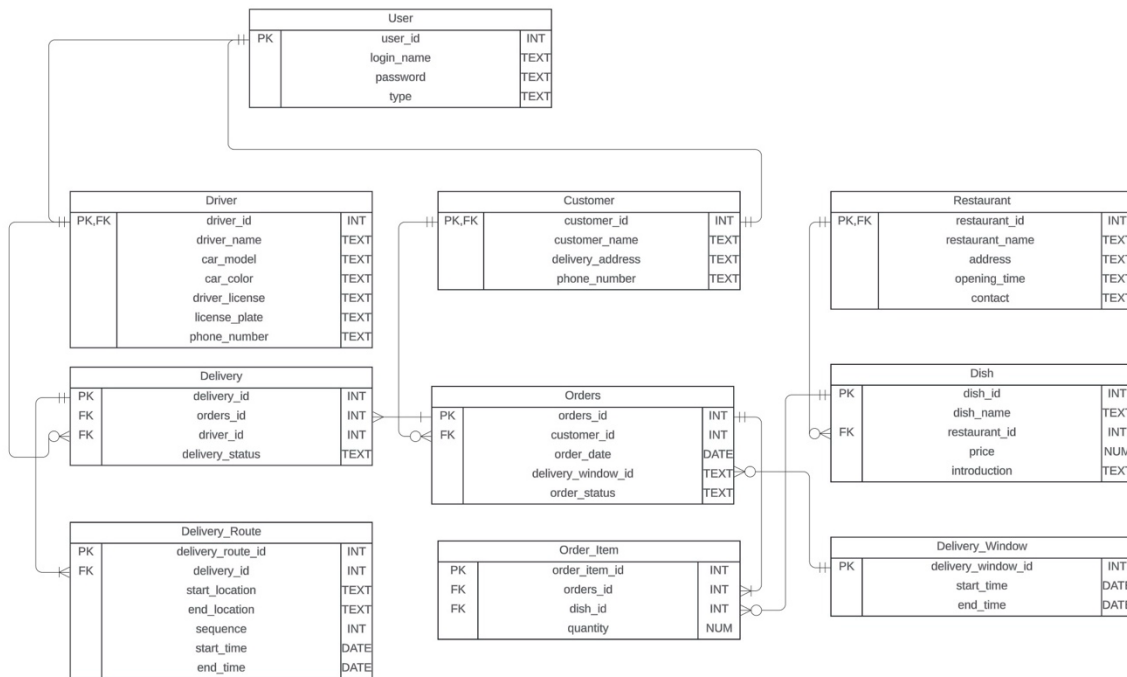


Relational Schema for Pre-prepared Dishes Ordering System

By CS5200 Fall 2024 semester Qiong Wu Project 1

ERD



Relational Schema

- User (user_id: INTEGER, login_name: TEXT, password: TEXT, type: TEXT)
- Customer (customer_id: INTEGER, customer_name: TEXT, delivery_address: TEXT, phone_number: TEXT)
- Restaurant (restaurant_id: INTEGER, restaurant_name: TEXT, restaurant_address: TEXT, opening_time: TEXT, contact: TEXT)
- Dish (dish_id: INTEGER, dish_name: TEXT, price: REAL, introduction: TEXT)
- Orders (orders_id: INTEGER, *customer_id*: INTEGER, order_date: TEXT, order_status: TEXT)

- Order_Item (order_item_id: INTEGER, *orders_id*: INTEGER, *dish_id*: INTEGER, quantity: INTEGER)
- Delivery (delivery_id: INTEGER, *orders_id*: INTEGER, *driver_id*: INTEGER, delivery_status: TEXT, delivery_date: DATE)
- Driver (driver_id: INTEGER, driver_name: TEXT, car_model: TEXT, car_color: TEXT, driver_license: TEXT, license_plate: TEXT, phone_number: TEXT)
- Delivery_Route (delivery_route_id: INTEGER, *delivery_id*: INTEGER, start_location: TEXT, end_location: TEXT, sequence: INTEGER, start_time: DATE, end_time: DATE)
- Delivery_Window (delivery_window_id, start_time: DATE, end_time: DATE)

Functional dependencies:

- **User:**
 - user_id -> login_name, password, type
 - user_id is a primary key
- **Customer:**
 - customer_id -> delivery_address, phone_number
 - customer_id is a primary key
- **Restaurant:**
 - restaurant_id -> restaurant_name, restaurant_address, opening_time, closing_time, contact
 - restaurant_id is a primary key
- **Dish:**
 - dish_id -> dish_name, price, introduction
 - dish_id is a primary key
- **Orders:**
 - orders_id -> customer_id, order_date, order_status
 - orders_id is a primary key
- **Order_Item:**
 - order_item_id -> orders_id, dish_id, quantity
 - (orders_id, dish_id) -> quantity
 - order_item_id is a primary key

- (orders_id, dish_id) is unique
- **Delivery:**
 - delivery_id -> orders_id, driver_id, delivery_status, delivery_route_id
 - delivery_id is a primary key
- **Driver:**
 - driver_id -> driver_name, car_model, car_color, driver_license, license_plate, phone
 - driver_id is a primary key
- **Delivery_Route:**
 - delivery_route_id -> start_location, end_location, sequence, start_time, end_time
 - delivery_route_id is a primary key
- **Delivery_Window:**
 - delivery_window_id -> start_time, end_time
 - delivery_window_id is a primary key